INCH-POUND
MIL-DTL-24643/73A
28 January 2010
SUPERSEDING
MIL-DTL-24643/73
9 October 2008

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -20 °C TO + 90 °C, 1000 VOLTS, TYPE LSTCF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction, Non-Watertight

First	-	Copper conductor, tin-coated, size 200, 550 strands of 24 AWG
Second	-	Optional separator.
Third	-	Thermoset insulation. Nominal wall thickness of 0.055 inch. Circuit identification shall be accomplished by solid-colored insulation or printed marking on the wire in accordance with table II.
Fourth	-	Three conductors cabled together with a lay in accordance with 3.4.4 of MIL-DTL-24643. Fillers may be used as necessary to form a firm well-rounded assembly.
Fifth	-	Optional binder.
Sixth	-	Cross-linked polyolefin jacket. Color black. Nominal wall thickness of 0.075 inch (see table I).

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TABLE I. Construction details.

Military part no. M24643/73	Type & size	AWG conductor size	Conductor resistance (ohms)	Cable jacket thickness (min. avg.) (inch)	Minimum bend radius (inch)	Overall diameter (inches)	
						(min.)	(max.)
-01UN	LSTCF-200	0000	0.054	0.065	9	1.700	1.800

TABLE II. Circuit identification.

Component	Identification
1	Black
2	White
3	Red

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic Electricals:

Conductor resistance (ohms/1000 feet at 25 °C, max.)	See table I
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	3000
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	500
Conductor continuity	No failure

Group A:

Visual and dimensional	No failure

Group B:

Cold bending cable	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 200 °C)	50
Physicals (unaged)	
Insulation	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150

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Jacket (cable)	
Tensile strength (lb/in^2, min.)	1300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (insulation) (Method 1 only; cycles, min.)	25
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (125±5 °C)	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50

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Smoke index, max.	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, max.	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – Type A (hardness, min.)	80
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSTCF	250

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-091)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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